SNA4DS cheatsheet

Automatically generated from the current snafun vignette tables and figures.

Contents

In this *cheatsheet* we summarize the main `R` functions that are used in the SNA4DS course. We will not explain the underlying concepts here, but refer you to the lectures, labs, and slides of the course for that.

The aim of this cheatsheet is that it provides you with an overview of the main functions you will need throughout the course. We hope that it can provide a useful reference for you, as you develop and apply your network analysis skills.

> **NOTE:**
>
> Most functions have multiple arguments. Our aim is in this cheatsheet not to show and discuss the various arguments that exist, because that would yield an unwieldy and very long document. Rather, we recommend you use your `R` skills and use the help function `?` and `help` and other approaches we teach you in this course to learn about the details of a specific function. If you still can't figure it out, contact us and we'll assist you.

Main packages in the SNA4DS course

Overview of igraph and network graph objects

There are two main packages for basic graph generation and manipulation: the `igraph` package and the `statnet` package. Actually, `statnet` is a suite of packages that work together. In this course, we will make use of several packages from the `statnet` suite.

The `igraph` package creates a graph object of type `igraph` . The `statnet` suite creates a graph object of type `network` . Both can generate graphs and do basic manipulation. The `igraph` package provides more mathematical graph functions, while the `statnet` suite provides many statistical models.

The snafun package

The `igraph` package and `statnet` suite are jointly very powerful, but they each require graph objects with specific structures. If you want to combine functions from `igraph` , `network` , and `sna` , you often end up converting between `igraph` , `network` , `matrix` , and edgelist representations.

Believe it or not, this is a pain and quite annoying.

THE `snafun` PACKAGE TO THE RESCUE!

- It provides a fairly consistent API, so you don't have to constantly re-learn different argument conventions.
- Most functions work on `igraph` , `network` , matrices, and edgelists/data frames.
- It lets you focus on the fun of network analysis rather than the friction of conversions and mismatched interfaces.

Creating a graph object

Graph creation

CREATE AN EMPTY NETWORK

snafun
```
snafun::create_empty_graph(n_vertices, directed = TRUE, graph = c("igraph", "network"))
```

igraph
```
igraph::make_empty_graph()
```

network
```
network::network.initialize(n)
```

CREATE A MANUAL (LITERAL) NETWORK

snafun
```
snafun::create_manual_graph(A -+ B -+ C, simplify = TRUE, graph = 'igraph')
```

igraph
```
igraph::graph_from_literal(A -+ B -+ C, simplify = TRUE)
```

network –

CREATE A RING NETWORK

snafun –

igraph
```
igraph::make_ring()
```

network –

CREATE A STAR NETWORK

snafun –

igraph
```
igraph::make_star()
```

network –

CREATE A RANDOM GRAPH WITH GIVEN DENSITY

snafun
```
snafun::create_random_graph(
    n_vertices = 10,
    strategy = "gnm",   # to fix the number of edges
    m = 27,             # number of edges required, resulting density = .3
    directed = TRUE,    # or FALSE
    graph = c("igraph", "network"))   # pick one

snafun::create_random_graph(
    n_vertices = 10,
```

[1] The output is a matrix object, not a 'network' object.

[2] This function is quite unintuitive and cumbersome

Graph creation

```
    strategy = "gnp",   # to fix the probability edges
    p = .3,             # probability for each edge, yields approx. density of .3
    directed = TRUE,    # or FALSE
    graph = c("igraph", "network"))
```

igraph

```
igraph::sample_gnm()
igraph::sample_gnp()

igraph::make_directed_graph(n = 10, edges = 27)
igraph::make_undirected_graph(n = 10, edges = 27)
```

network[1]

```
# 10 vertices, density on average .3
sna::rgraph(n = 10, m = 1, tprob = 0.30, mode = 'digraph')

# 10 vertices, 27 edges (ie. density = .3)
sna::rgnm(1, 10, m = 27)
```

CREATE A RANDOM GRAPH WITH GIVEN DYAD CENSUS

snafun

```
create_census_graph(n_vertices = 10, mut = 8, asym = 25, null = 12,
                                        method = 'exact',
                                        graph = 'igraph')
```

igraph –

network[1]

```
# 5 networks, each with 10 vertices, and 8 M, 25 A, and 12 N dyads
sna::rguman(n = 5, nv = 10, mut = 8, asym = 25, null = 12,
  method = "exact")
```

CREATE A RANDOM GRAPH WITH GIVEN COMMUNITY STRUTURE

snafun

```
snafun::create_community_graph(
  communitySizes = c(10, 20, 30),
  p_intra = c(0.3, 0.2, 0.3),
  p_inter = 0.2,
  p_del = 0,
  graph = 'igraph')
```

igraph –

network –

CREATE A GRAPH WITH GIVEN COMPONENTS

snafun

```
snafun::create_components_graph(
```

[1] The output is a matrix object, not a 'network' object.

[2] This function is quite unintuitive and cumbersome

Graph creation

```
  n_vertices,
  directed = FALSE,
  membership = NULL,
  graph = 'igraph')
```

igraph –

network –

CREATE RANDOM BIPARTITE GRAPH

snafun

```
snafun::create_bipartite(
  n_type1,                        # number of vertices of type 1
  n_type2,                        # number of vertices of type 2
  strategy = c("gnp", "gnm"),
  p,                              # probability of each cross-type edge
  m,                              # number of cross-type edges
  directed = FALSE,
  mode = c("out", "in", "all"),
  graph = c("igraph", "network")
)
```

igraph

```
igraph::sample_bipartite_gnp(n1, n2, p)
```

network[2]

```
network::network.bipartite()
```

CREATE GRAPH OBJECT FROM INPUT DATA

snafun

```
#  `x` can be:
#   - an edgelist (in data.frame format)
#   - an incidence matrix (in matrix format)--for a bipartite graph
#   - an adjacency matrix (in matrix format)
#   - `vertices`  can be a data.frame containing vertex attributes

snafun::to_igraph(x, bipartite = FALSE, vertices = NULL)

snafun::to_network(x, bipartite = FALSE, vertices = NULL)
```

igraph

```
# from an adjacency matrix
igraph::graph_from_adjacency_matrix(adjmatrix,
  mode = c("directed", "undirected", "max", "min", "upper", "lower", "plus"),
  weighted = NULL, diag = TRUE, add.colnames = NULL, add.rownames = NA)

igraph::make_graph(edges, ..., n = max(edges), isolates = NULL, directed = TRUE,
  dir = directed, simplify = TRUE)
```

[1] The output is a matrix object, not a 'network' object.

[2] This function is quite unintuitive and cumbersome

Graph creation

```
# from an adjacency list
igraph::graph_from_adj_list(adjlist, mode = c("out", "in", "all", "total"),
  duplicate = TRUE)

# from an edgelist
igraph::graph_from_edgelist(el, directed = TRUE)

# from a data.frame
igraph::graph_from_data_frame(d, directed = TRUE, vertices = NULL)

# from a bipartite incidence / biadjacency matrix (in matrix format)
igraph::graph_from_biadjacency_matrix(incidence, directed = FALSE,
  mode = c("all", "out", "in", "total"), multiple = FALSE,
  weighted = NULL, add.names = NULL)
```

network

```
# x can be an adjacency matrix (as a matrix), an incidence matrix (as a matrix),
# or an edgelist (as a data.frame)
network::network(x, vertex.attr = NULL, vertex.attrnames = NULL, directed = TRUE,
  hyper = FALSE, loops = FALSE, multiple = FALSE, bipartite = FALSE, ...)

# if x is a data.frame
network::as.network(x, directed = TRUE, vertices = NULL, hyper = FALSE,
  loops = FALSE, multiple = FALSE, bipartite = FALSE,
  bipartite_col = "is_actor", ...)

# if x is a matrix
network::as.network(x, matrix.type = NULL, directed = TRUE, hyper = FALSE,
  loops = FALSE, multiple = FALSE, bipartite = FALSE, ignore.eval = TRUE,
  names.eval = NULL, na.rm = FALSE, edge.check = FALSE, ...)
```

READ DATA FROM A UCINET FILE

snafun

```
snafun::read_ucinet(file,
  graph = c("igraph", "network", "matrix", "edgelist"),
  directed = c("auto", "directed", "undirected"))
```

igraph	–
network	–

BUILD AN EDGELIST / NODELIST FROM RAW DATA

snafun

```
snafun::make_edgelist(names = NULL, attribute = NULL)
snafun::make_nodelist(names = NULL, attribute = NULL)
```

igraph	–

[1] The output is a matrix object, not a 'network' object.

[2] This function is quite unintuitive and cumbersome

Graph creation	
network	–

[1] The output is a matrix object, not a 'network' object.

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Additional graph creation functions

After creating a graph, you can also extend it with helper functions such as `snafun::add_vertices()` and `snafun::add_edges()` .

Converting between graph classes

Graph object conversion

CONVERT TO AN ADJACENCY MATRIX

igraph	`igraph::as_adjacency_matrix(g, sparse = FALSE)`
network	`network::as.sociomatrix(g)` `network::as.matrix.network(g)`

CONVERT TO AN ADJACENCY LIST

igraph	`igraph::as_adj_list(g)`
network	–

CONVERT TO AN EDGE LIST

igraph	`igraph::as_edgelist(g)` `igraph::as_data_frame(g)`

network	`network::as.data.frame.network(g)`
	`network::as.edgelist(g)`
	
	`# if you want to include edge weights`
	`sna::as.edgelist.sna(g)`

MAKE THE NETWORK DIRECTED

snafun	–
igraph	`igraph::as_directed(g)`
network	–

MAKE THE NETWORK UNDIRECTED

snafun	`# from an adjacency matrix as input`
	`snafun::to_symmetric_matrix(g,`
	`  rule = c("weak", "mutual", "out", "in", "average", "max", "min"),`
	`  na.rm = TRUE)`

igraph	`igraph::as_undirected(g)`
	
	`# for example`

	Graph object conversion
	`igraph::as_undirected(g, mode = 'collapse', edge.attr.comb = list(weight = 'sum'))`

network	`# from an adjacency matrix as input`
	`sna::symmetrize(g)`
	
	`# for example`
	`sna::symmetrize(g, rule = "strong")`

PROJECT A BIPARTITE GRAPH	

snafun	`snafun::make_projection(x, which = "both")`
	
	`# or request a single partition`
	`snafun::make_projection(x, which = "row")`
	`snafun::make_projection(x, which = "column")`

igraph	`igraph::bipartite_projection(g)`
network	–
CONVERT TO A LINE GRAPH	
snafun	–
igraph	`igraph::make_line_graph(g)`
network	–
DICHOTOMIZE A MATRIX (MAKE BINARY)	
snafun	`# values below 'min' become 0, the rest become 1` `snafun::to_binary_matrix(mat, min)`
igraph	–
network	–

Additional graph creation functions

The cheatsheet below focuses on the most common creation routes. The package also contains helper functions for adding vertices and edges after creation.

Convert between various formats

Using the consistent functions of **snafun**

	OUTPUT			
INPUT	**edgelist**	**matrix**	**igraph**	**network**
edgelist	`to_edgelist()`	`to_matrix()`	`to_igraph()`	`to_network()`
matrix	`to_edgelist()`	`to_matrix()`	`to_igraph()`	`to_network()`
igraph	`to_edgelist()`	`to_matrix()`	`to_igraph()`	`to_network()`
network	`to_edgelist()`	`to_matrix()`	`to_igraph()`	`to_network()`

Conversion includes bipartite graphs.
Check the arguments for the various options.

Manipulating the graph object

Manipulate the graph object

PRINT THE GRAPH OBJECT

snafun	`snafun::print(x)`
igraph[1]	`igraph::print.igraph(x)`
network[1]	`network::print.network(x)`

CHECK CHARACTERISTICS OF THE GRAPH OBJECT

snafun	`snafun::has_edge_attributes(x)`
	`snafun::has_vertex_attributes(x)`
	`snafun::has_vertex_attribute(x, attrname)`
	`snafun::has_edge_attribute(x, attrname)`
	`snafun::has_vertexnames(x)`
	`snafun::has_loops(x)`
	`snafun::is_bipartite(x)`
	`snafun::is_connected(x, rule = c("weak", "strong"))`
	`snafun::is_directed(x)`
	`snafun::is_network(x)`
	`snafun::is_igraph(x)`
	`snafun::is_signed(x)`
	`snafun::is_weighted(x)`

igraph	`# various functions, including`
	`igraph::any_loop(g)`
	`igraph::any_multiple(g)`
	`igraph::is_bipartite(graph)`
	`igraph::is_connected(graph, mode = c("weak", "strong"))`
	`igraph::is_directed(graph)`
	`igraph::is_igraph(graph)`

[1] Often, but certainly not always, it will also work if you just use `print(x)`

[2] These functions have many additional arguments

Manipulate the graph object

```
igraph::is_named(graph)
igraph::is_simple(graph)
igraph::is_weighted(graph)
```

network

```
# various functions, including:
network::has.loops(x)
network::is.bipartite(x)
network::is.directed(x)
network::is.multiplex(x)
network::is.network(x)
sna::is.connected(g, connected = "strong")
```

ACCESS THE VERTICES AND EDGES

snafun –

igraph

```
igraph::V(x)              # vertices
igraph::E(x)              # edges
```

network –

EXTRACT VERTEX / EDGE / GRAPH ATTRIBUTES

snafun

```
snafun::extract_vertex_attribute(x, name)
snafun::extract_vertex_names(x)             # specific to access the vertex names attribute
snafun::extract_edge_attribute(x, name)
snafun::extract_graph_attribute(x, name)
```

igraph

```
igraph::vertex_attr(graph, name, index = V(graph))
igraph::V(graph)$attributename

igraph::edge_attr(graph, name, index = E(graph))
igraph::E(graph)$attributename

igraph::graph_attr(graph, name)
```

network[2]

```
network::get.vertex.attribute(x, attrname)
network::get.edge.attribute(x, attrname)
network::get.network.attribute(x, attrname)
```

igraph –

network –

[1] Often, but certainly not always, it will also work if you just use `print(x)`

[2] These functions have many additional arguments

Manipulate the graph object

EXTRACT THE ID OF AN EDGE

snafun
```
snafun::extract_edge_id(object, ego, alter, edgelist)
```

EXTRACT VERTEX NAMES

snafun
```
snafun::extract_vertex_names(x)
```

igraph
```
igraph::V(x)$name
igraph::vertex_attr(x, name = "name")
```

network
```
network::get.vertex.attribute(x, "vertex.names")
network::network.vertex.names(x)
```

SET VERTEX NAMES

snafun
```
snafun::add_vertex_names(x, value)
```

igraph
```
igraph::V(x)$name &lt;- value
igraph::set_vertex_attr(x, name = "name", value = value)
```

network
```
network::set.vertex.attribute(x, "vertex.names", value)
network::network.vertex.names(x) &lt;- value
```

EXTRACT INTERNAL VERTEX IDS

snafun
```
snafun::extract_vertex_ids(x)
```

igraph –

network –

MAKE A MATRIX FROM A VERTEX ATTRIBUTE

snafun
```
# e.g. to build covariate matrices for ERGM edgecov()/absdiff()
snafun::make_matrix_from_vertex_attribute(x, name,
  measure = c("absdiff", "diff", "sum", "max", "min",
              "mean", "sender", "receiver", "equal"))
```

igraph –

[1] Often, but certainly not always, it will also work if you just use `print(x)`

[2] These functions have many additional arguments

Manipulate the graph object

network	–

LIST VERTEX / EDGE / GRAPH ATTRIBUTES

snafun	`snafun::list_vertex_attributes(x)`
	`snafun::list_edge_attributes(x)`
	`snafun::list_graph_attributes(x)`

igraph	`igraph::vertex_attr_names(graph)`
	`igraph::edge_attr_names(graph)`
	`igraph::graph_attr_names(graph)`

network	`network::list.vertex.attributes(x)`
	`network::list.edge.attributes(x)`
	`network::list.network.attributes(x)`

EXTRACT ALL VERTEX ATTRIBUTES INTO A DATA.FRAME

snafun	`snafun::extract_all_vertex_attributes(g)`
igraph	–
network	–

ADD VERTEX / EDGE / GRAPH ATTRIBUTES

snafun	`# very flexible and powerful functions to add attributes`
	`# in several ways, with the same function`
	`snafun::add_edge_attributes(object, attr_name, value, edgelist, overwrite = FALSE)`
	
	`snafun::add_vertex_attributes(x, attr_name = NULL, value)`
	`snafun::add_graph_attribute(x, attr_name = NULL, value)`

igraph	`igraph::set_edge_attr(g, "name", value)`
	`igraph::E(g)$name &lt;- value`
	
	`igraph::set_vertex_attr(g, "name", value)`
	`igraph::V(g)$name &lt;- value`
	
	`igraph::set_graph_attr(g, "name", value)`
	`g$name &lt;- value`

[1] Often, but certainly not always, it will also work if you just use `print(x)`

[2] These functions have many additional arguments

Manipulate the graph object

network	`network::set.edge.attribute(g, "new_name", value, e = seq_along(g$mel))`
	`network::set.edge.value(g, "new_name", value, e = seq_along(g$mel))`
	
	`network::set.network.attribute(g, "new_name", value)`
	
	`network::set.vertex.attribute(g, "new_name", value, v = seq_len(network::network.size(g)))`

REMOVE VERTEX / EDGE / GRAPH ATTRIBUTES

snafun	`snafun::remove_edge_attribute(x, attr_name)`
	`snafun::remove_edge_weight(x)  # remove edge weights`
	
	`snafun::remove_vertex_attribute(x, attr_name)`
	`snafun::remove_vertex_names(x)   # specific for the vertex name attribute`
	
	`snafun::remove_graph_attribute(x, attr_name)`

igraph	`igraph::delete_edge_attr(graph, name)`
	`igraph::delete_vertex_attr(graph, name)`
	`igraph::delete_graph_attr(graph, name)`

network	`network::delete.edge.attribute(x, attrname, ...)`
	`network::delete.vertex.attribute(x, attrname, ...)`
	`network::delete.network.attribute(x, attrname, ...)`

ADD VERTICES OR EDGES

snafun	`snafun::add_vertices(x, vertices)` `snafun::add_edges(x, edges)`
igraph	`igraph::add_vertices(graph, nv, ..., attr = list())` `igraph::add_edges(graph, edges, ..., attr = list())`

network	`network::add.vertices(x, nv, vattr = NULL, last.mode = TRUE, ...)`
	`network::add.edge(x, tail, head, names.eval = NULL, vals.eval = NULL, edge.check = FALSE, ...)`
	`network::add.edges(x, tail, head, names.eval = NULL, vals.eval = NULL, ...)`

REMOVE VERTICES OR EDGES

snafun	`snafun::remove_vertices(x, vertices)`

[1] Often, but certainly not always, it will also work if you just use `print(x)`

[2] These functions have many additional arguments

Manipulate the graph object

igraph	`igraph::delete_vertices(graph, v)` `igraph::delete_edges(graph, edges)`
network	`network::delete.edges(x, eid)` `network::delete.vertices(x, vid)`

CONTRACT VERTICES INTO ONE

snafun	`snafun::contract_vertices(g, vertices, method = c("min", "max", "union", "add"), attr = NULL, new_name = "set", out_as_igraph = TRUE)`
igraph	–
network	–

EXTRACT A SUBSET FROM THE GRAPH

snafun	`# single function to do it all` `snafun::extract_subgraph(x, v_to_keep, e_to_keep)`

igraph	`# subset based on vertices`
	`igraph::induced_subgraph(g, vids = theVerticesYouWantToKeep)`
	
	`# subset based on edges`
	`igraph::subgraph_from_edges(g, eids = theEdgesYouWantToKeep)`

network	`# subset based on vertices`
	`network::get.inducedSubgraph(g, v = theVerticesYouWantToKeep)`
	
	`# subset based on edges`
	`network::get.inducedSubgraph(g, eid = theEdgesYouWantToKeep)`

EXTRACT EGONETS FROM THE GRAPH

snafun	`snafun::extract_egonet(x, vertices = NULL, order = 1, type = c("all", "out", "in"))`
igraph	`igraph::make_ego_graph(g, order = 1, nodes = V(graph), mode = c("all", "out", "in"), mindist = 0)`
network	`sna::ego.extract(dat, ego = NULL, neighborhood = c("combined", "in", "out"))`

[1] Often, but certainly not always, it will also work if you just use `print(x)`

[2] These functions have many additional arguments

Manipulate the graph object

CLEAN UP THE GRAPH

snafun

```
snafun::extract_isolates(x, names = TRUE, loops = FALSE)
snafun::remove_isolates(x, loops = FALSE)
snafun::remove_loops(x)
snafun::extract_loops(x)          # which edges are loops
snafun::extract_loops_vertex(x)   # which vertices carry a loop
snafun::has_multiple_edges(x)
snafun::extract_multiple_edges(x)
snafun::remove_multiple_edges(x)
```

igraph

```
igraph::simplify(g)

# for example
igraph::simplify(g, remove.multiple = TRUE,
    remove.loops = TRUE, edge.attr.comb = list(weight = 'max'))
```

network

```
sna::isolates(g, diag=FALSE)
```

UNION OF GRAPHS

snafun

```
snafun::make_union(net1, net2, net3, byname = 'auto')
```

igraph

```
igraph::union(net1, net2, net3)
```

network –

INTERSECTION OF GRAPHS

snafun

```
snafun::extract_intersection(net, net2, net3, byname = 'auto', keep.all.vertices = TRUE)
```

igraph

```
igraph::intersection(net, net2, net3)
```

network –

[1] Often, but certainly not always, it will also work if you just use `print(x)`

[2] These functions have many additional arguments

Graph level indices

Explore the graph

SUMMARIZE THE NETWORK

snafun	`snafun::g_summary(x, directed = TRUE)`
igraph	–
network	–

COUNT THE NUMBER OF VERTICES

snafun	`snafun::count_vertices(x)`
igraph	`igraph::vcount()`
network	`network::network.size()`

COUNT THE NUMBER OF EDGES

snafun	`snafun::count_edges(x)`
igraph	`igraph::ecount(x)` `igraph::gsize(x)`
network	`network::network.edgecount(x)`

DENSITY

snafun	`snafun::g_density(x, loops = FALSE)`
igraph	`igraph::edge_density(g)`

network	`# preferable for bipartite graphs`
	`network::network.density(g)`
	
	`# preferable for valued graphs`

[1] This function is more general than degree assortativity only: it can be computed on any vertex attribute (via `attrname` or `values`).

[2] igraph and sna only calculate `Freeman` centralization (snafun does both `Freeman` and `sd`)

[3] `\$res` or `\$vector` return the centrality scores

	Explore the graph
	`sna::gden(g)`
RECIPROCITY	
snafun	`snafun::g_reciprocity(x)`
igraph	`igraph::reciprocity(g)`
network	`sna::grecip(g, measure = 'edgewise')`
TRANSITIVITY	
snafun	`snafun::g_transitivity(x)`
igraph	`igraph::transitivity(g, type = 'global')`
network	`sna::gtrans(g, mode = 'digraph', measure = 'weak', use.adjacency = TRUE)`
MEAN DISTANCE BETWEEN VERTICES	
snafun	`snafun::g_mean_distance(x)`
igraph	`igraph::mean_distance(g, directed = TRUE, unconnected = TRUE)`
network	–
DEGREE DISTRIBUTION	
snafun	`snafun::g_degree_distribution(x, mode = c("out", "in", "all"), type = c("density", "count"), cumulative = FALSE, loops = FALSE, digits = 3)`
igraph	`igraph::degree_distribution(graph, cumulative = FALSE, mode = 'out')`

[1] This function is more general than degree assortativity only: it can be computed on any vertex attribute (via `attrname` or `values`).

[2] igraph and sna only calculate `Freeman` centralization (snafun does both `Freeman` and `sd`)

[3] `\$res` or `\$vector` return the centrality scores

	Explore the graph
network	–
DYAD CENSUS	
snafun	`snafun::count_dyads(x, echo = TRUE)`
igraph	`igraph::dyad_census(g)`
network	`sna::dyad.census(g)`
TRIAD CENSUS	
snafun	`snafun::count_triads(x, echo = TRUE)`
igraph	`igraph::triad_census(g)`
network	`sna::triad.census(g, mode = 'digraph')`
DEGREE ASSORTATIVITY	
snafun[1]	`snafun::g_assortativity(x, attrname = NULL, values = NULL, vertices = NULL, directed = TRUE, normalized = TRUE)`
igraph	`igraph::assortativity_degree(g, directed = TRUE)`
network	–
DIAMETER	
snafun	`snafun::g_diameter(x, directed = is_directed(x), unconnected = TRUE)`

igraph	`igraph::diameter(g, directed = TRUE, unconnected = TRUE)`
	
	`# what is the vertex pair with the longest geodesic`
	`igraph::farthest_vertices(g)`

network	–

[1] This function is more general than degree assortativity only: it can be computed on any vertex attribute (via `attrname` or `values`).

[2] igraph and sna only calculate `Freeman` centralization (snafun does both `Freeman` and `sd`)

[3] `\$res` or `\$vector` return the centrality scores

Explore the graph

RADIUS

snafun
```
snafun::g_radius(x, mode = c("all", "out", "in"))
```

igraph
```
igraph::radius(graph, mode = c("all", "out", "in", "total"))
```

network –

COMPACTNESS

snafun
```
snafun::g_compactness(x, mode = c("out", "in", "all"))
```

igraph –

network –

CENTRALIZATION

snafun
```
# a single function to calculate centralization (either `Freeman` or `sd`)
# on a wide range of centrality indices

snafun::g_centralize(x, measure = "betweenness",
  directed = TRUE, mode = c("all", "out", "in"),
  k = 3, damping = 0.85, normalized = TRUE,
  method = c("freeman", "sd"))
```

igraph[2,3]
```
# general function for Freeman centralization
igraph::centralize(scores, theoretical.max = 0, normalized = TRUE)

# betweenness centralization
igraph::centr_betw(g, directed = TRUE)

# closeness centralization
igraph::centr_clo(g, mode = 'out', normalized = FALSE)

# degree centralization
igraph::centr_degree(g, mode = 'all')

# eigenvector centralization
igraph::centr_eigen(g, directed = TRUE)
```

network[2]
```
# general function for Freeman centralization,
# include any function that calculates vertex centrality
```

[1] This function is more general than degree assortativity only: it can be computed on any vertex attribute (via `attrname` or `values`).

[2] igraph and sna only calculate `Freeman` centralization (snafun does both `Freeman` and `sd`)

[3] `\$res` or `\$vector` return the centrality scores

Explore the graph

```
sna::centralization(g, FUN, mode = 'digraph', normalize=TRUE, ...)
```

MIXING MATRIX

snafun
```
snafun::make_mixingmatrix(x, attrname, by_edge = FALSE, loops = has_loops(x))
```

igraph –

network
```
network::mixingmatrix(object, attrname, useNA = "ifany", expand.bipartite = FALSE)
```

CORRELATION BETWEEN TWO GRAPHS

snafun
```
snafun::g_correlation(g1, g2, diag = FALSE)
```

igraph –

network
```
sna::gcor(g1, g2, mode = 'graph')
```

GRAPH EFFICIENCY

snafun
```
snafun::g_efficiency(g, diag = FALSE)
```

igraph –

network
```
sna::efficiency(g, diag = FALSE)
```

SECRECY INDEX

snafun
```
snafun::g_secrecy(g, type = 0, p = 0.25, digits = 3)
```

igraph –

network –

VULNERABILITY: ATTACK

snafun
```
snafun::g_vuln_attack(g, mode = c("all", "out", "in"),
  weight = NA, k = 10, digits = 4)
```

igraph –

network –

[1] This function is more general than degree assortativity only: it can be computed on any vertex attribute (via `attrname` or `values`).

[2] igraph and sna only calculate `Freeman` centralization (snafun does both `Freeman` and `sd`)

[3] `\$res` or `\$vector` return the centrality scores

Explore the graph

VULNERABILITY: EFFICIENCY		
snafun		`snafun::g_vuln_efficiency(g, method = c("harmonic", "sum"), mode = c("all", "out", "in"), weight = NA, disconnected = c("size", "max", "infinite"), digits = 3)`
igraph	–	
network	–	
VULNERABILITY: PATHS		
snafun		`snafun::g_vuln_paths(g, mode = c("all", "out", "in"), weight = NULL, digits = 3)`
igraph	–	
network	–	

[1] This function is more general than degree assortativity only: it can be computed on any vertex attribute (via `attrname` or `values`).

[2] igraph and sna only calculate `Freeman` centralization (snafun does both `Freeman` and `sd`)

[3] `\$res` or `\$vector` return the centrality scores

Components and communities

Components and communities

COUNT THE NUMBER OF COMPONENTS

snafun	`snafun::count_components(x, type = c("weak", "strong"))`
igraph	`igraph::count_components(graph, mode = c("weak", "strong"))`
network	`sna::components(dat, connected = "weak")`

EXTRACT THE COMPONENTS

snafun	`snafun::extract_components(x, type = c("weak", "strong"))`
igraph	`igraph::decompose(graph, mode = c("weak", "strong"))`
network	–

EXTRACT COMPONENT MEMBERSHIP

snafun	`snafun::extract_component_membership(x, type = c("weak", "strong"))`
igraph	`igraph::components(graph)$membership`
network	–

EXTRACT CUT VERTICES (ARTICULATION POINTS)

snafun	`snafun::extract_cut_vertices(x, names = TRUE)`
igraph	`igraph::articulation_points(graph)`
network	–

EXTRACT BRIDGES

snafun	`snafun::extract_bridges(x)`
igraph	`igraph::bridges(graph)`
network	–

FAST-GREEDY COMMUNITY DETECTION

Components and communities

snafun	`snafun::extract_comm_fastgreedy(x, weights = NA, modularity = TRUE, merges = TRUE, membership = TRUE)`
igraph	`igraph::cluster_fast_greedy(g, merges = TRUE, modularity = TRUE, membership = TRUE, weights = NULL)`
network	–

GIRVAN-NEWMAN COMMUNITY DETECTION

snafun	`snafun::extract_comm_girvan(x, weights = NA, directed = TRUE, modularity = TRUE, edge.betweenness = FALSE, bridges = FALSE, merges = TRUE, membership = TRUE)`
igraph	`igraph::cluster_edge_betweenness(g, weights = NULL, directed = TRUE, edge.betweenness = TRUE, merges = TRUE, bridges = TRUE, modularity = TRUE, membership = TRUE)`
network	–

LOUVAIN COMMUNITY DETECTION

snafun	`snafun::extract_comm_louvain(x, weights = NA, resolution = 1)`
igraph	`igraph::cluster_louvain(graph, weights = NULL, resolution = 1)`
network	–

WALKTRAP COMMUNITY DETECTION

snafun	`snafun::extract_comm_walktrap(x, weights = NA, steps = 4, modularity = TRUE, merges = TRUE, membership = TRUE)`
igraph	`igraph::cluster_walktrap(g, weights = NULL, steps = 4, merges = TRUE, modularity = TRUE, membership = TRUE)`
network	–

MERGE COMMUNITY MEMBERSHIP

snafun	`snafun::merge_membership(coms, merges)`
igraph	–
network	–

COMMUNITY MEMBERSHIP

Components and communities

snafun	`snafun::extract_comm_membership(coms)`
igraph	`igraph::membership(coms)`
network	–

NUMBER OF COMMUNITIES

snafun	`snafun::count_communities(coms)`
igraph	`length(coms)`
network	–

COMMUNITY SIZES

snafun	`snafun::extract_comm_sizes(coms)`
igraph	`igraph::sizes(coms)`
network	–

COMMUNITY MODULARITY

snafun	`snafun::extract_comm_modularity(coms)`
igraph	`igraph::modularity(coms)`
network	–

COMMUNITY CROSSING TIES

snafun	`snafun::extract_comm_crossing(coms, graph)`
igraph	`igraph::crossing(coms, graph)`
network	–

EVALUATE COMMUNITIES

snafun	`snafun::evaluate_communities(coms, graph, sil_width = FALSE)`

Components and communities

igraph	–
network	–

PLOT SILHOUETTE WIDTHS FOR COMMUNITIES

snafun	evls <- snafun::evaluate_communities(coms, graph) plot(evls)
igraph	–
network	–

PLOT COMMUNITIES

snafun	snafun::plot_communities(coms, graph)
igraph	plot(coms, graph)
network	–

ADD COMMUNITY MEMBERSHIP TO GRAPH

snafun	snafun::add_comm_membership(graph, coms, attr = "community")
igraph	–
network	–

ADD COMMUNITY CROSSING TO GRAPH

snafun	snafun::add_comm_crossing(graph, coms, attr = "crossing")
igraph	–
network	–

SUMMARIZE A COMMUNITY PARTITION

snafun	snafun::summarize_communities(x, graph = NULL)
igraph	–
network	–

PLOT A COMMUNITY DENDROGRAM

snafun	snafun::plot_comm_dendrogram(x, labels = NULL, hang = 0.1)
igraph	igraph::plot_dendrogram(coms)

Components and communities

network	–

WHICH DETECTION ALGORITHM WAS USED

snafun	`snafun::extract_comm_algorithm(x)`
igraph	`igraph::algorithm(coms)`
network	–

EXTRACT THE COMMUNITY MERGES

snafun	`snafun::extract_comm_merges(x)`
igraph	`igraph::merges(coms)`
network	–

Community results returned by `snafun::extract_comm_*()` are compatible with the wider `igraph` community API. You can inspect memberships, sizes, crossings, modularity, and evaluate the detected communities with `snafun::evaluate_communities()` .

Vertex-level indices

Explore the vertices	
DEGREE	
snafun	`snafun::v_degree(x, vids = NULL, mode = c("all", "out", "in"), loops = FALSE, rescaled = FALSE)`
igraph	`igraph::degree(g, mode = 'out')`
network	`sna::degree(g, gmode = 'digraph', cmode = 'outdegree')`
BETWEENNESS	
snafun	`snafun::v_betweenness(x, vids = NULL, directed = TRUE, rescaled = FALSE)`
igraph	`igraph::betweenness(g, directed = TRUE)`
network	`sna::betweenness(g, gmode = 'digraph', cmode = 'directed')`
FLOW BETWEENNESS	
snafun	`snafun::v_flow(x, vids = NULL, mode = c("rawflow", "normflow", "fracflow"), rescaled = FALSE)`
igraph	–
network	`sna::flowbet(g, gmode = 'digraph', cmode = 'rawflow')`
BONACICH POWER CENTRALITY	
snafun	`snafun::v_power(x, vids = NULL, exponent = 1, rescaled = FALSE)`
igraph	`igraph::power_centrality(g)`
network	`sna::bonpow(g, gmode = 'digraph')`
CLOSENESS CENTRALITY	

[1] These bridge measures need a community assignment. `communities` can be the output of `snafun::extract_comm_*()`, a membership vector, or a named list of communities.

[2] For these shortest-path bridge measures, positive weights are translated to distances via `1/weight`. Zero and negative weights are ignored in the path search; use `weights = NA` for purely topological paths.

Explore the vertices

snafun	`snafun::v_closeness(x, vids = NULL, mode = c("all", "out", "in"), rescaled = FALSE)`
igraph	`igraph::closeness(g, mode = 'all')`
network	`sna::closeness(g, gmode = 'digraph', cmode = 'directed')`

BRIDGE STRENGTH

snafun[1]	`snafun::v_bridge_strength(`
	`  x,`
	`  communities,`
	`  use_communities = NULL,`
	`  type = c("all", "out", "in"),`
	`  absolute = TRUE,`
	`  rescaled = FALSE`
	`)`

igraph	–
network	–

BRIDGE EXPECTED INFLUENCE

snafun[1]	`snafun::v_bridge_expected_influence(`
	`  x,`
	`  communities,`
	`  use_communities = NULL,`
	`  type = c("all", "out", "in"),`
	`  rescaled = FALSE`
	`)`

igraph	–
network	–

BRIDGE EXPECTED INFLUENCE (2-STEP)

snafun[1]	`snafun::v_bridge_expected_influence2(`
	`  x,`
	`  communities,`
	`  use_communities = NULL,`
	`  type = c("all", "out", "in"),`
	`  rescaled = FALSE`
	`)`

[1] These bridge measures need a community assignment. `communities` can be the output of `snafun::extract_comm_*()`, a membership vector, or a named list of communities.

[2] For these shortest-path bridge measures, positive weights are translated to distances via `1/weight`. Zero and negative weights are ignored in the path search; use `weights = NA` for purely topological paths.

	Explore the vertices
igraph	–
network	–
BRIDGE CLOSENESS	

snafun[2]	`snafun::v_bridge_closeness(`
	`  x,`
	`  communities,`
	`  use_communities = NULL,`
	`  mode = c("out", "in", "all"),`
	`  weights = NULL,`
	`  rescaled = FALSE`
	`)`

igraph	–
network	–
BRIDGE BETWEENNESS	

snafun[2]	`snafun::v_bridge_betweenness(`
	`  x,`
	`  communities,`
	`  use_communities = NULL,`
	`  directed = NULL,`
	`  weights = NULL,`
	`  rescaled = FALSE`
	`)`

igraph	–
network	–
HARMONIC CENTRALITY	
snafun	`snafun::v_harmonic(x, vids = NULL, mode = c("all", "out", "in"), rescaled = FALSE)`
igraph	`igraph::harmonic_centrality(g, vids = V(graph),` `  mode = c("out", "in", "all"), weights = NULL)`
network	–
STRESS CENTRALITY	
snafun	`snafun::v_stress(x, vids = NULL, directed = TRUE, rescaled = FALSE)`
igraph	–

[1] These bridge measures need a community assignment. `communities` can be the output of `snafun::extract_comm_*()`, a membership vector, or a named list of communities.

[2] For these shortest-path bridge measures, positive weights are translated to distances via `1/weight`. Zero and negative weights are ignored in the path search; use `weights = NA` for purely topological paths.

Explore the vertices

network	`sna::stresscent(g, gmode = 'digraph', cmode = 'directed')`

ECCENTRICITY

snafun	`snafun::v_eccentricity(x, vids = NULL, mode = c("all", "out", "in"), rescaled = FALSE)`
igraph	`igraph::eccentricity(g, mode = 'all')`
network	–

EIGENVECTOR CENTRALITY

snafun	`snafun::v_eigenvector(x, directed = TRUE, rescaled = FALSE)`
igraph	`igraph::eigen_centrality(g, directed = TRUE, scale = FALSE)$vector`
network	`sna::evcent(g, gmode = 'digraph', rescale=FALSE)`

PAGE RANK

snafun	`snafun::v_pagerank(x, vids = NULL, damping = 0.85, directed = TRUE, rescaled = FALSE)`
igraph	`igraph::page_rank(g, vids = V(graph), directed = TRUE, damping = 0.85, weights = NULL)`
network	–

GEO K-PATH

snafun	`# without using weights`
	`snafun::v_geokpath(x, vids = NULL, mode = c("all", "out", "in"),`
	`  k = 3, rescaled = FALSE)`
	
	`# if weights are to be used`
	`snafun::v_geokpath_w(x, vids = NULL, mode = c("all", "out", "in"),`
	`  weights = NULL, k = 3)`

igraph	–
network	–

SHAPLEY CENTRALITY

[1] These bridge measures need a community assignment. `communities` can be the output of `snafun::extract_comm_*()`, a membership vector, or a named list of communities.

[2] For these shortest-path bridge measures, positive weights are translated to distances via `1/weight`. Zero and negative weights are ignored in the path search; use `weights = NA` for purely topological paths.

Explore the vertices

snafun	`snafun::v_shapley(x, add.vertex.names = FALSE, vids = NULL, rescaled = FALSE)`
igraph	–
network	–

DISTANCES TO AND FROM A VERTEX

snafun	`snafun::v_distance(x, mode = c("all", "out", "in"), weights = NULL, count_unnconnected = FALSE)`
igraph	`igraph::distances(g, v = igraph::V(g), mode = 'out')`
network	`sna::geodist(g)$gdist`

VERTEX TRANSITIVITY (LOCAL CLUSTERING)

snafun	`snafun::v_transitivity(x, vids = NULL, isolates = c("nan", "zero"))`
igraph	`igraph::transitivity(g, type = 'local')`
network	–

BOTTLENECK CENTRALITY

snafun	`snafun::v_bottleneck(graph, mode = c("all", "out", "in"), vids = igraph::V(graph), n = 4)`
igraph	–
network	–

DIFFUSION CENTRALITY

snafun	`snafun::v_diffusion(x, vids = NULL, T = NULL, rescaled = FALSE)`
igraph	–
network	–

FRAGMENTATION

snafun	`snafun::v_fragment(x, vids = NULL, M = Inf, binary = FALSE,`

[1] These bridge measures need a community assignment. `communities` can be the output of `snafun::extract_comm_*()`, a membership vector, or a named list of communities.

[2] For these shortest-path bridge measures, positive weights are translated to distances via `1/weight`. Zero and negative weights are ignored in the path search; use `weights = NA` for purely topological paths.

		Explore the vertices
		large = TRUE, rescaled = FALSE)
igraph	–	
network	–	
VERTEX SECRECY		
snafun		snafun::v_secrecy(g, type = 1, p = 0.25, digits = 3)
igraph	–	
network	–	
WHO ARE THE NEIGHBORS OF A VERTEX		
snafun		snafun::extract_neighbors(x, vertex, type = c("out", "in", "all"))
igraph		igraph::neighbors(g, 'Jane', mode = 'out') # all options igraph::neighbors(graph, v, mode = c('out', 'in', 'all', 'total'))
network		network::get.neighborhood(g, 1, 'out') # all options network::get.neighborhood(x, v, type = c('out', 'in', 'combined'), na.omit = TRUE)

[1] These bridge measures need a community assignment. `communities` can be the output of `snafun::extract_comm_*()`, a membership vector, or a named list of communities.

[2] For these shortest-path bridge measures, positive weights are translated to distances via `1/weight`. Zero and negative weights are ignored in the path search; use `weights = NA` for purely topological paths.

Dyad-level indices

Explore the dyads

SHORTEST PATH FOR A GIVEN SET OF VERTICES

snafun	`snafun::extract_all_shortest_paths(x, from, to = NULL, mode = c("out", "in", "all"), weights = NULL)`
igraph	`igraph::all_shortest_paths(g, from = IDofVertex, to = igraph::V(g), mode = 'out')`
network	–

GEODESIC LENGTHS[1]

snafun	`snafun::d_distance(x, mode = c("all", "out", "in"))`
igraph	`igraph::distances(g, mode = 'out')`
network	`sna::geodist(g, count.paths = TRUE)$counts`

STRUCTURAL EQUIVALENCE

snafun	`snafun::d_structural_equivalence(x, weights = NA, digits = 3, suppressWarnings = TRUE)`
igraph	–
network	`sna::sedist(g, method = "correlation", mode ="digraph", diag=FALSE)`

REGULAR EQUIVALENCE

snafun	`snafun::d_regular_equivalence(x, weights = NA, digits = 3)`
igraph	–
network	`1 - sna::redist(g, method = "catrege", mode = "digraph", diag = FALSE)`

EDGE BETWEENNESS

snafun	`snafun::d_betweenness(x, directed = TRUE, weights = NA)`
igraph	`igraph::edge_betweenness(g, directed = FALSE)`

[1] The output is a table with an entry per vertex pair

Explore the dyads

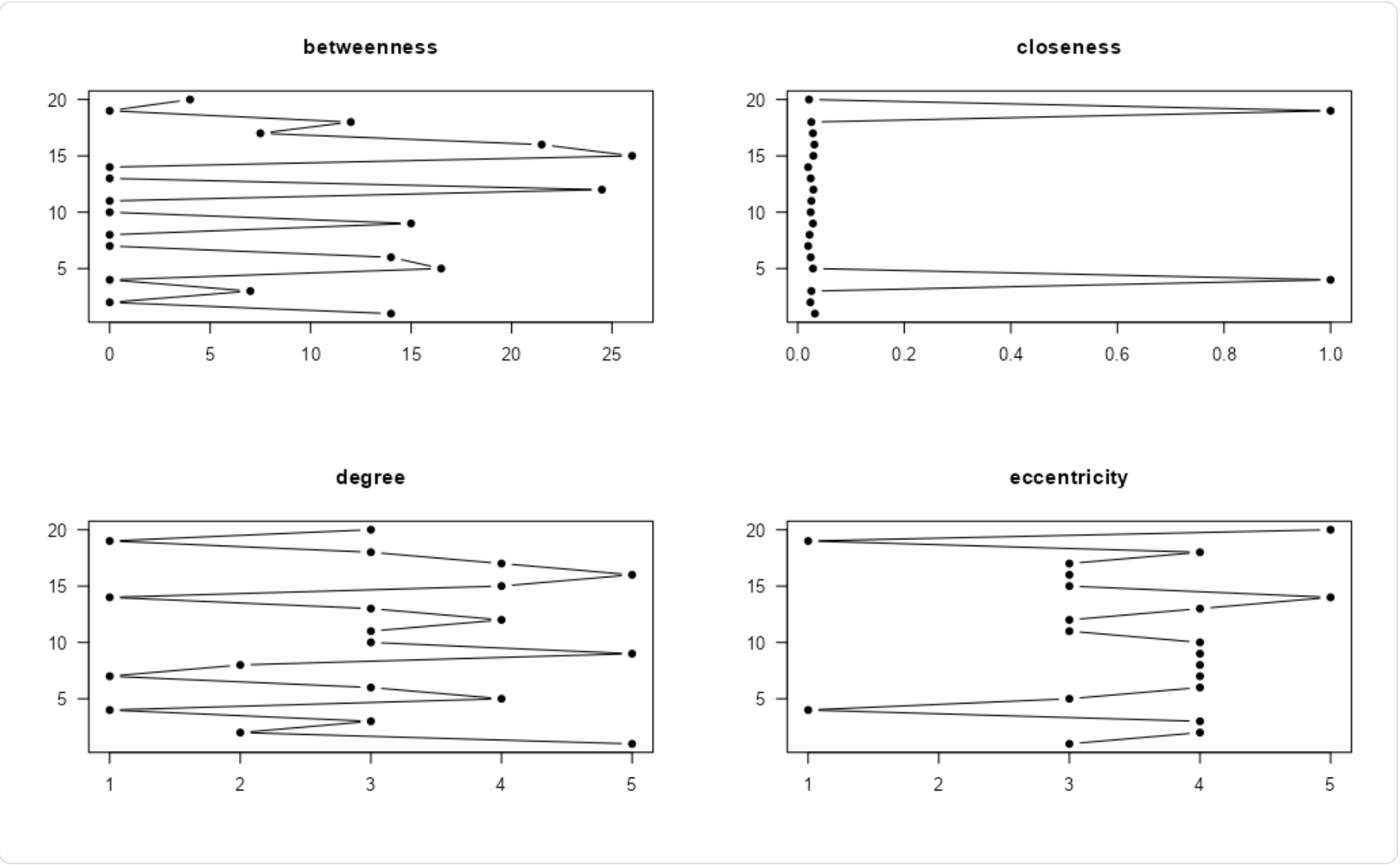
network	–

1 The output is a table with an entry per vertex pair

Plotting

Plotting in snafun

The `snafun` package provides S3 plotting methods, so a plain call to `plot(x)` gives a quick plot regardless of whether `x` is an `igraph` or `network` object. The package also includes a dedicated centrality comparison plot through `snafun::plot_centralities()` .



Example output from snafun::plot_centralities().

Statistical models

Overview table

Statistical network models

When	Which approach	Function
Dependent vertex attribute explained by a network weight matrix and a matrix of covariates	Network autocorrelation model	`snafun::stat_nam`
A statistic on a single network	Conditional Uniform Graph test	`snafun::stat_cug`
Association between two networks	QAP	`snafun::stat_qap_cor`
A valued dependent network explained by one or more explanatory networks	QAP linear model	`snafun::stat_qap_lm`
A binary dependent network explained by one or more explanatory networks	QAP logistic model	`snafun::stat_qap_logit`
A binary or valued dependent network explained by a set of endogenous and exogenous variables	Exponential random graph models	`ergm::ergm`

Network autocorrelation models

`snafun::stat_nam(formula, data, W, model = c("lag", "error", "combined"))` -- fit a network autocorrelation model, where `W` is the (row-normalized) network weight matrix.

`snafun::stat_nam_summary(x, correlation = TRUE, R2 = TRUE, digits = 3)` -- a full summary (coefficients, correlations, R2) of a fitted NAM.

`snafun::plot_nam(x)` -- diagnostic plots (e.g. residuals vs. fitted) for a fitted NAM.

Conditional Uniform graphs (CUG)

`snafun::stat_cug(x, FUN, cmode = c("size", "edges", "dyad.census"), reps = 1000)` -- test the statistic `FUN` on a single network against random graphs conditioned on `cmode` .

CUG (and QAP below) inference is permutation-based; the table below shows how to randomly permute a network.

Permuting the network

RANDOMLY PERMUTE THE ORDER OF THE VERTICES	
snafun	–
igraph	`igraph::permute(graph = x, permutation = sample(igraph::vcount(x)))`
network	`sna::rmperm(x)`

QAP test

`snafun::stat_qap_cor(x, y, controls = NULL, reps = 1000)` -- test the association between two networks (optionally controlling for others).

QAP linear regression

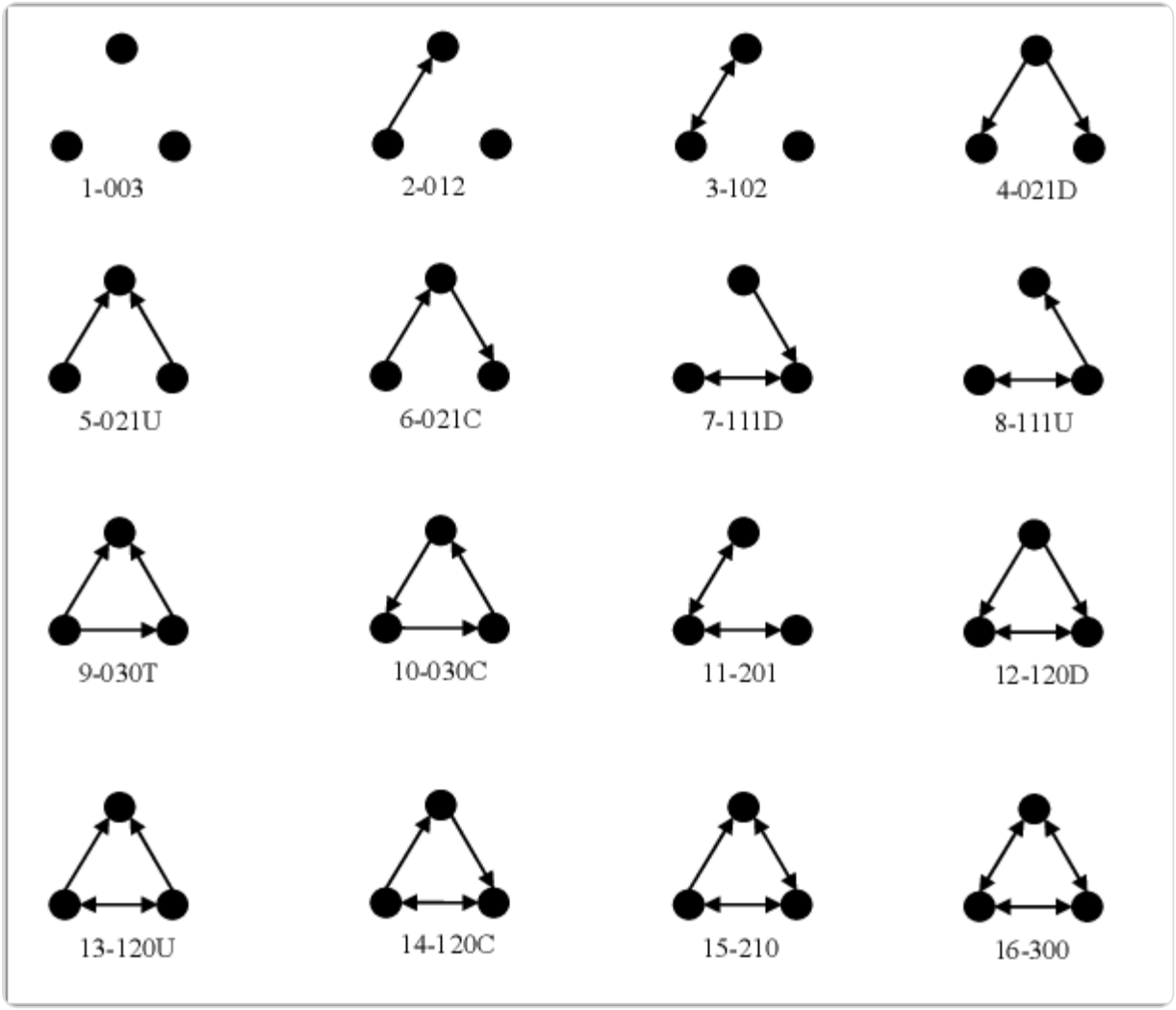
`snafun::stat_qap_lm(y, x, reps = 1000)` -- a valued dependent network `y` explained by one or more predictor networks `x` .

QAP logistic regression

`snafun::stat_qap_logit(y, x, reps = 1000)` -- same idea, but for a binary dependent network y .

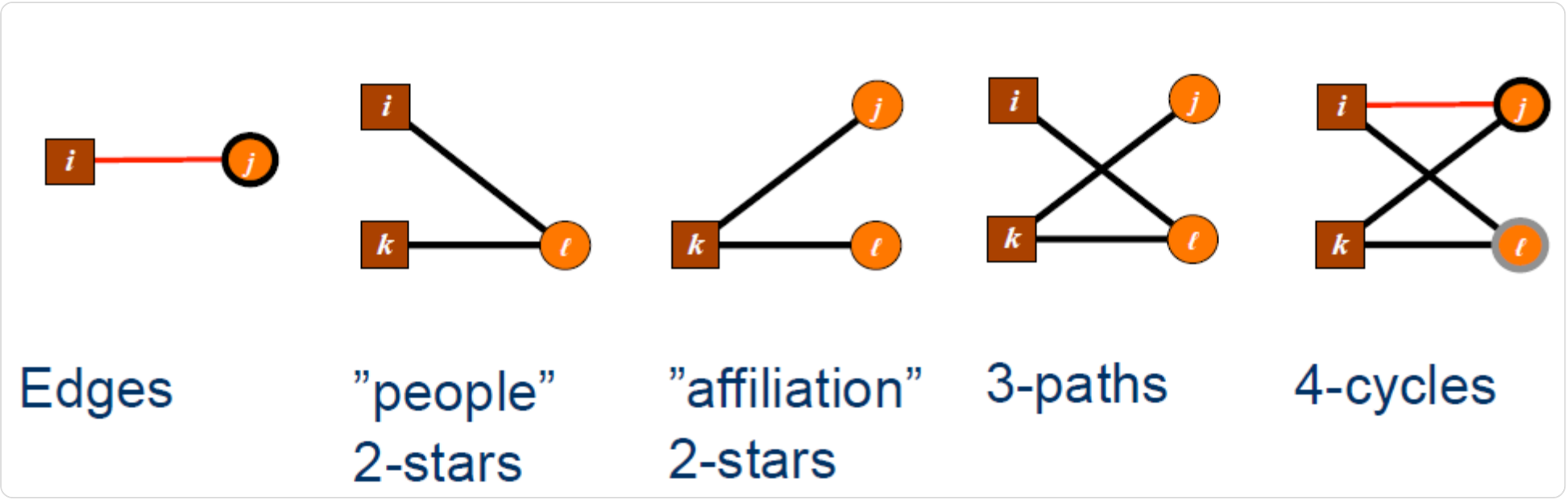
Terms classification for every Exponential Random Graph Model (ERGM)

The static figures below are used in the course to explain common ERGM term families.



Triad census categories used in ERGM teaching.

Bipartite ERGMs



Overview of common bipartite ERGM terms.

ERGM goodness of fit and effect interpretation

`snafun::stat_plot_gof(gof)` -- plot an already-computed goodness-of-fit object of a fitted ergm or btergm.

`snafun::stat_plot_gof_as_btergm(m, silent = FALSE, verbose = TRUE)` -- compute AND plot the goodness of fit for a fitted model `m` via the btergm engine (works for both ergm and btergm).

`snafun::stat_ef_int(m, type = "odds")` -- translate the effects of a fitted ergm into odds ratios (or probabilities).

Temporal networks

Network generation and manipulation

In this course, temporal network work mostly relies on `networkDynamic` , `tsna` , and `ndtv` . The `snafun` API now also supports several active-attribute accessors for `networkDynamic` objects.

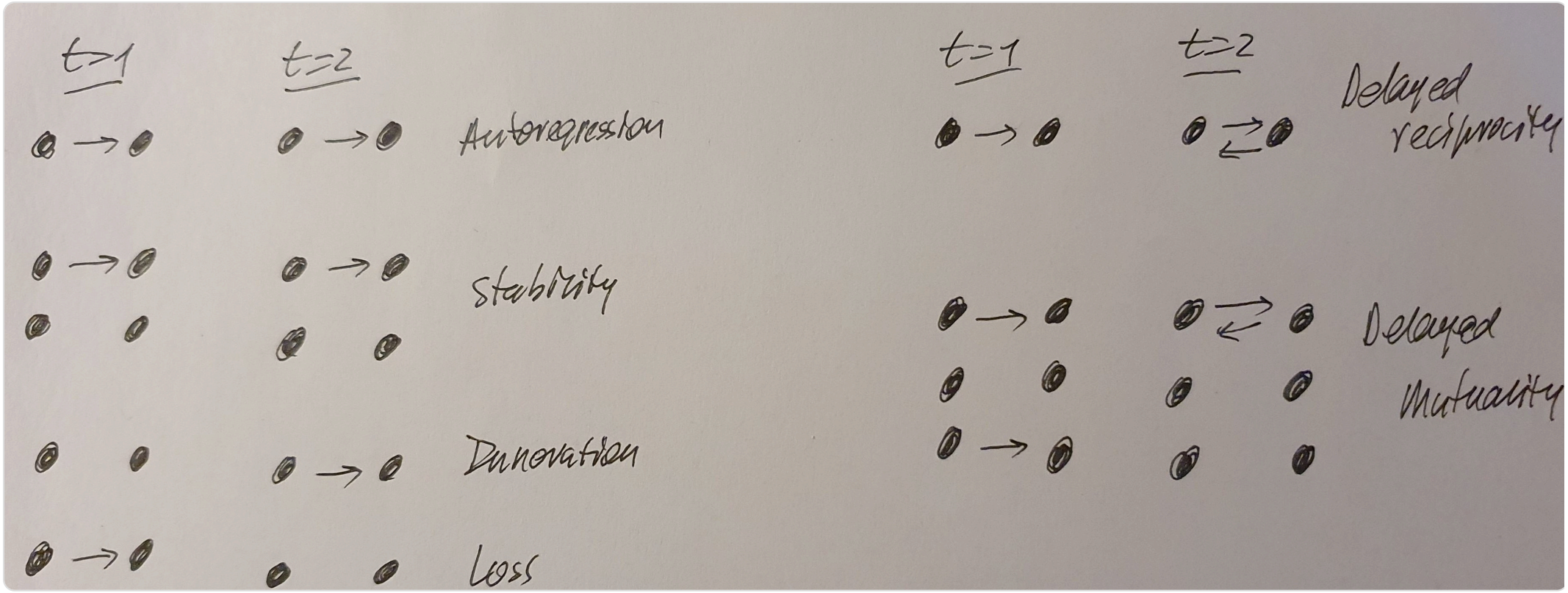
Network measures and descriptives

Temporal descriptives often focus on durations, edge formations, dissolutions, and time-varying ERGM terms. The snafun helpers below operate on `networkDynamic` objects.

Temporal network descriptives	
PLOT NETWORK SNAPSHOTS OVER TIME (FILMSTRIP)	
snafun	`snafun::plot_network_slices(x, number = 9, start = NULL, end = NULL)`
COUNT EDGES PRESENT OVER A TIME INTERVAL	
snafun	`snafun::count_edges_in_interval(x, start = NULL, end = NULL, number = 30)`
COUNT UNIQUE EDGES OVER A TIME INTERVAL	
snafun	`snafun::count_unique_edges_in_interval(x, start = NULL, end = NULL, number = 30, directed = network::is.directed(x))`

ERGM for temporal networks

Temporal ERGMs (`btergm`) extend ERGMs with memory, delayed-reciprocity and time-covariate terms. The figure and table below summarize the common temporal terms.



Overview of common temporal ERGM terms.

Temporal effects for the ERGM

	meaning	btergm
memory		
Positive autoregression	Previous existing edges persist in a next network	`memory(type = "autoregression", lag = 1)`
Dyadic stability	Both previous existing and non-existing ties are carried over to the current network	`memory(type = "stability", lag = 1)`
Edge innovation	A non-existing previous tie becomes existent in the current network	`memory(type = "innovation", lag = 1)`
Edge loss	An existing previous tie is dissolved in the current network	`memory(type = "loss", lag = 1)`
delayed reciprocity		
reciprocity	if node j is tied to node i at t = 1, does this lead to a reciprocation of that tie back from i to j at t = 2?	`delrecip(mutuality = FALSE, lag = 1)`
mutuality	if node j is tied to node i at t = 1, does this lead to a reciprocation of that tie back from i to j at t = 2 AND if i is not tied to j at t = 1, will this lead to j not being tied to i at t = 2? This captures a trend away from asymmetry.	`delrecip(mutuality = TRUE, lag = 1)`
time covariates		
time effect per se	Test for a specific trend (linear or non-linear) for edge formation	`timecov(transform = function(t) t)`
Time effect of a covariate	Interaction effect to test whether the importance of a covariate increases or decreases over time	`timecov(x, transform = function(t) t)`

These are ERGM model *terms* used **inside** the model formula, e.g. `... ~ edges + memory(type = 'stability') + delrecip()`. `timecov()` is an exported btergm function, while `memory()` and `delrecip()` are model terms (documented under btergm's tergm-terms help), so they are not called as `btergm::memory()`.

Participation shifts

P-shift	Example
Turn receiving	
AB-BA	John talks to Mary, then Mary replies.
AB-B0	John talks to Mary, then Mary addresses the group.
AB-BY	John talks to Mary, then Mary talks to Irene.
Turn claiming	
A0-X0	John talks to the group, then Frank talks to the group.
A0-XA	John talks to the group, then Frank talks to John.
A0-XY	John talks to the group, then Frank talks to Mary.
Turn usurping	
AB-X0	John talks to Mary, then Frank talks to the group.
AB-XA	John talks to Mary, then Frank talks to John.
AB-XB	John talks to Mary, then Frank addresses Mary.
AB-XY	John talks to Mary, then Frank addresses Irene.
Turn continuing	
A0-AY	John talks to the group, then addresses Mary.
AB-A0	John talks to Mary, then makes a remark to the group.
AB-AY	John talks to Mary then to Irene.

Note: The initial speaker is always labeled A, and the initial target B, unless the group is addressed (or the target was ambiguous), in which case the target is 0. Then the shift is summarized in the form $[\text{speaker}_1][\text{target}_1]$-$[\text{speaker}_2][\text{target}_2]$, with A or B appearing after the hyphen only if the initial speaker or target serves in one of these two positions after the shift. When the speaker after the shift is someone other than A or B, X is used, and when the target after the shift is someone other than A, B, or the group, Y is used.

Participation-shift categories used in the temporal networks part of the course.